

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD - 402 103
End Semester Examination, May 2018

Branch: B. Pharmacy

Subject with Subject Code: Pharmaceutical Organic Chemistry
[BP202T]

Semester: II

Marks: 75

Date: 24 / 05 / 2018

Time: 3 Hrs.

Instructions:

1] All Questions are compulsory.

2] Structures to the right indicate full marks.

Que. No. 1 Attempt the following multiple choice questions.

(20x1 = 20)

1. The peroxide effect is observed with only
 - a) HI
 - b) HBr
 - c) HCl
 - d) HF
2. The addition of unsymmetrical reagent to unsymmetrical alkenes occurs such a way that the negative part of the adding molecule goes to the carbon atom of the double bond which carries less number of hydrogen atom is.....
 - a) Anti-Markovnikov rule
 - b) Peroxide effect
 - c) Markovnikov rule
 - d) Saytzeff rule
3. The one step elimination reaction which follows second order kinetics and doesn't involve rearrangement is called as....
 - a) E1 reaction
 - b) E2 reaction
 - c) Ecb reaction
 - d) None of above
4. The IUPAC name of isobutane is
 - a) 2-methyl propane
 - b) 2-methyl propene
 - c) 2-methyl propyne
 - d) None of above
5. In conjugated dienes, double bonds are separated by
 - a) one single sigma bond
 - b) two single sigma bond
 - c) three single sigma bond
 - d) more than one single sigma bond
6. Which one is saturated compound?
 - a) Propylene
 - b) Propane
 - c) Ethylene
 - d) Formamide
9. The carbon atom involved in the double bond of alkenes is
 - a) sp hybridized
 - b) sp^2 hybridized
 - c) sp^3 hybridized
 - d) None of these
10. Which statement about carbonyl compound is NOT true.
 - (a) The carbonyl group is SP^2 hybridized
 - (b) The bond angles among the three atoms attached to the carbonyl carbon are 120 degree
 - (c) The three atoms attached to the carbonyl carbon form non planer geometry
 - (d) The carbonyl group containing carbon-oxygen double bond

11. For aldol condensation,.....is required in aldehyde and ketone

(a) alpha- hydrogen	(b) beta –hydrogen
(c) gamma- hydrogen	(d) OH- group
12. The final product of cannizaro reactions is.....

(a) Aldehyde	(b) Ketone
(c) Alcohol and salt of carboxylic acid	(d) Hydrazone
13. The appearance of a silver mirror in tollen's test indicates the presence of.....

(a) an aldehyde	(b) a ketone
(c) an alcohol	(d) an alkenes
14. In SN^1 reaction, rate of reaction is depending upon reactants

(a) one	(b) two
(c) three	(d) four
15. The IUPAC name of $Br-CH_2 - CH_2 - CH_2 - CHO$ is

a) 1 -Bromobutanal	b) 4-Bromobutanol
c) 1 –Bromobutane	d) 4-Bromobutanal
16. Alkyl halides undergo

a) Electrophilc substitution reaction	b) Electrophilc addition reaction
c) Nucleophilc substitution reaction	d) Nucleophilc addition reaction
17. Which compound reacts most rapidly by an SN^1 mechanism?

a) Methyl chloride	b) Isopropyl chloride
c) Ethyl chloride	d) t-Butyl chloride
18. The nature of aliphatic amines are....

a) Acidic	b) Basic
c) Neutral	d) partly acidic
19. test is used as distinguishing test for alcohols.

a) Lucas	b) Amino
c) Carbohydrate	d) Ritzs
20. Which of the following is strongest acid?

a) Formic acid	b) Acetic acid
c) Chloroacetic acid	d) Propionic acid

Que. No. 2 Attempt any two of the following:

(10x2 = 20)

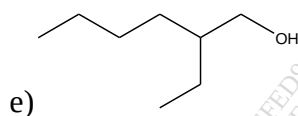
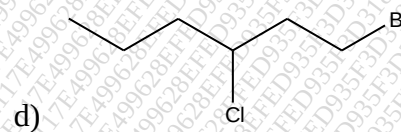
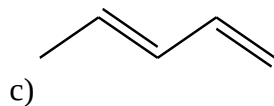
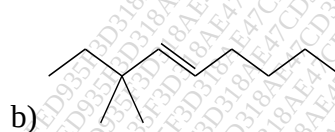
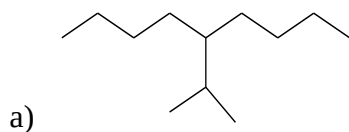
- A)** What is elimination reaction? Give detail account of E1 and E2 reactions of alkenes.
- B)** What are carboxylic compounds? Explain acidity of carboxylic acid and effect of substitution on acidity of carboxylic acid.
- C)** What is Nucleophilic Substitution Reaction? Explain SN^1 and SN^2 reactions in detail.

Que. No 3. Attempt any seven of the following:

(5x7 = 35)

A) What is isomerism? Explain its types with suitable example.

B) Give IUPAC name to the following compounds



C) What are Aldehydes and Ketones? Give any one method of preparation of each.

D) Explain the reaction with mechanism:

a) Aldol condensation

b) Cannizzaro reaction

E) Draw structures and write uses of

a. Formaldehyde

b. Ethyl alcohol

c. Ethyl chloride

d. Chloroform

e. Chloral hydrate

F) Explain basicity of aliphatic amines.

G) Explain any two methods of preparation of alcohols.

H) Draw the structure of following compounds

a. 5-ethyl-3-methylbutane

b. 2,4-dimethyl-2-pentene

c. 1-bromo-3-methylbutane

d. 3-methyl-1-butanol

e. 3-bromopentanal

I) Write a note on Diels – Alder reaction.

□□□□□



**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE-RAIGAD-402103**

Winter Semester Examination December 2019

Branch- B.Pharm

Subject:- Pharmaceutical Organic Chemistry-I (BP202T)

Date-05/12/2019

Sem-II

Marks: 75

Time -3 hrs

Instructions:

- i) All questions are compulsory
- ii) Figures to the right indicates full marks

Q.No. 1 Attempt the following multiple choice question (20×1=20)

1. IUPAC stands for----
 - a) International Union of pure & applied chemistry
 - b) Indian union of pure & application chemistry
 - c) Indian United of pure & applied chemistry
 - d) None of these
2. Alcohols react with PCl_3 to give
 1. Alkanes b) Alkenes c) Acid d) Alkyl halide
3. S_N^1 reaction occurs through the intermediate formation of
 - a) Carbocation b) Carbanion c) Free radical d) all of these
4. The acidity of Carboxylic acid is----- by addition of electron releasing alkyl group
 - a) Increases b) Decreases c) Remains constant d) None of these
5. Primary amines can be synthesized by
 - a) Reduction of Nitroalkanes b) Reduction of Nitriles
 - c) Reduction of oximes d) All of these
6. Cannizzaro reaction is given by
 - a) Aldehydes b) Ketones c) Both a & b d) None of these
7. The general formula for alkenes is
 - a) $\text{C}_n\text{H}_{2n-1}$ b) $\text{C}_n\text{H}_{2n+2}$ c) C_nH_{2n} d) None of these
8. ---- hybridization is observed in alkanes
 - a) Sp^2 b) Sp^3 c) $\text{Sp}^2\text{-Sp}^3$ d) $\text{Sp}^2\text{-Sp}^2$
9. IUPAC name of $\text{CH}_3\text{-CH(CH}_3\text{)-CH}_2\text{-CH(CH}_3\text{)-CH}_3$ is---
 - a) Dimethylpentane b) Methylpentane
 - c) 2,4 pentane d) 2,4-Dimethylpentane

10. Alcohols react with ammonia at 400°C to give mixture of----
 a) 1° & 3° amine b) $1^{\circ}, 2^{\circ}, 3^{\circ}$ amines c) 1° & 2° amine d) 1° & 3° amine
11. When the rate of an elimination reaction depends only on concentration of a substrate the reaction is of
 a) Second order b) First order c) Third order d) None of these
12. In Tertiary alkyl halides the attack of nucleophile is takes place through --- mechanism
 a) SN^1 b) SN^2 c) Either SN^1 or SN^2 d) None of these
13. ----- containing α Hydrogen undergo self addition in presence of base to give aldol condensation reaction
 a) Aldehydes b) Ketones c) both a & b d) Alcohol
14. Addition of Grignard reagent to aldehyde & ketone followed by hydrolysis gives
 a) Alkyl halide b) carboxylic acid c) Alkanes d) Alcohols
15. Amines are derivative of
 a) Ammonia b) Methane c) Acetic acid d) None of these
16. General formula for alcohol is
 a) $\text{C}_n\text{H}_{2n+1}$ b) $\text{C}_n\text{H}_{2n-1}$ c) C_nH_{2n} d) None of these
17. Identify the correct statement for alkanes
 a) It is known as paraffins b) In alkanes C-C bonds are single
 c) Alkanes have general formula $\text{C}_n\text{H}_{2n+2}$ d) All of above
18. Acetic acid is having molecular formula----
 a) CH_3COOH b) $\text{CH}_3\text{CH}_2\text{COOH}$ c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$ d) None of these
19. Aliphatic amines are ----in nature
 a) Acidic b) Basic c) Neutral d) All of above
20. ---- is a saturated compound
 a) Ethane b) Ethylene c) Propylene d) Propyne

Q .No. 2 Attempt any Two of following

($10 \times 2 = 20$)

1. Write note on reaction mechanism of $\text{E1}, \text{E2}$ elimination reaction along with difference between two
2. a) How will you distinguish between $1^{\circ}, 2^{\circ}, 3^{\circ}$ alcohols
 b) Write note on Diels-Alder reaction

3. a) Explain the basicity of aliphatic amines

b) Discuss the mechanism of Markownikoff's addition to alkenes with example

Q.No. 3 Attempt any SEVEN of following question

(5×7=35)

1. Draw the structure & write uses of

a) Chloroform b) Formic acid c) Ethanol d) Iodoform e) Ethane

2. Explain effect of substituent's on acidity of carboxylic acid.

3. Give the methods of preparation of alkanes.

4. Explain mechanism of halogenations of alkanes.

5. Write a note on Aldol condensation reaction.

6. Draw the following functional group

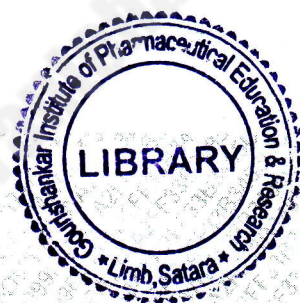
a) Nitro b) amino c) Halo d) Hydroxy e) Methyl

7. Explain the mechanism involved in Perkin reaction.

8. Differentiate between $SN1$ & $SN2$ reaction.

9. Explain Sp^3 hybridization of alkanes with example.

Paper End



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Summer 2022

Course: B. Pharmacy

Sem: II

Subject Name: Pharmaceutical Organic Chemistry I

Subject Code: BP202T

Max Marks: 75

Date: 29/08/2022

Duration: 3.45 Hr.

Instructions –

1. All questions are compulsory
2. Answers to MCQs should be written in full sentences
3. Draw diagrams / figures wherever necessary
4. Figures to right indicate full marks

Q. 1. Solve the following Multiple Choice Questions (MCQs) = $20 \times 1 = 20$ (All questions are compulsory)

i) The IUPAC name of the compound $\text{Cl-CH}_2\text{-CH}_2\text{-CH}_2\text{-OH}$ is.....

- | | |
|---------------------|----------------------|
| a) 1-chloropropanal | b) 3-chloropropanol |
| c) 3-chloropropane | d) 1-chloropropanone |

ii) Halogenation of alkanes takes place by mechanism.

- | | |
|-----------------|------------------|
| a) free radical | b) carbanion |
| c) carbocation | d) none of these |

iii) In elimination reaction of alkyl halides, the major product is that alkene in which double bond is more highly substituted, is the

- | | |
|-----------------------|---------------------------|
| a) Markovnikov's rule | b) Hoffman rule |
| c) Saytzeff rule | d) Antimarkovnikov's rule |

iv) The elimination reaction in which rate depends upon concentration of both substrate and base is

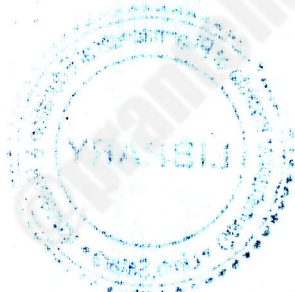
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| a) E1 reaction | b) E2 reaction |
| c) SN1 | d) none of these |

v) In addition of hydrogen halides to alkenes, peroxide effect is observed for

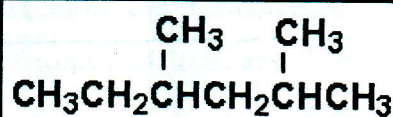
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|--------|---------------------|
| a) HCl | b) HBr |
| c) HI | d) all of the above |

vi) At low temperature addition to conjugated diene gives

- | | |
|-----------------|-----------------|
| a) 1,1 addition | b) 1,2 addition |
|-----------------|-----------------|



- c) 1,3 addition
d) 1,4 addition
- vii) Which of the following is not a nucleophile?
- a) $C_2H_5O^-$
b) SCN^-
c) OH^-
d) CH_3^+
- viii) Select the correct IUPAC name for



- a) 1,1,3-trimethylpentane
b) 2,4-dimethylhexane
c) 3,5-dimethylhexane
d) 3,5,5-trimethylpentane
- ix) What is the substitution product of the following reaction?
- $$\begin{array}{c} CH_3CHCH_3 \\ | \\ Br \end{array} + NaOH \longrightarrow$$
- a) 2-propanol
b) ethanol
c) 1-propanol
d) propane
- x) The nucleophilic substitution reaction in which rate of reaction depends upon concentration of only one reactant and rearrangement is possible is known as
- a) E1 reaction
b) SN_1 reaction
c) E2 reaction
d) SN_2 reaction
- xi) Lucas test is used to distinguish different types of
- a) alcohols
b) aldehydes
c) ketones
d) amines
- xii) Reduction of aldehyde gives
- a) Primary alcohol
b) Secondary alcohol
c) Tertiary alcohol
d) Carboxylic acid
- xiii) The major product of the following reaction is



- a) 1-propanol
b) 2-propanol
c) Propyne
d) Isopropyl chloride
- xiv) Reduction of ketone gives

- a) primary alcohol b) secondary alcohol
c) tertiary alcohol d) carboxylic acid
- xv) Addition of water to aldehyde gives.....
a) acetal b) hydrate
c) cyanohydrin d) oxime
- xvi) A mixture of benzaldehyde and formaldehyde on heating with aqueous NaOH solution gives
a) benzyl alcohol + sodium formate b) sodium benzoate + methanol
c) benzyl alcohol + methanol d) sodium benzoate + sodium formate
- xvii) The reaction of carboxylic acids with alcohols catalyzed by conc. H_2SO_4 is called ____
a) Dehydration b) Saponification
c) Esterification d) Neutralization
- xviii) Primary amines can be distinguished from secondary and tertiary amines by reacting with which of the following?
a) Chloroform and alcoholic KOH b) Methyl iodide
c) Chloroform alone d) Zinc dust
- xix) n-pentane, isopentane and neopentane are
a) functional isomers b) Metamers
c) chain isomers d) tautomers
- xx) The Hinsberg reagent is
a) benzenesulphonyl chloride b) diethyl oxalate
c) zinc chloride d) none of these

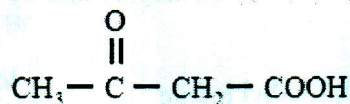
Q. 2. Solve the following questions. = $2 \times 10 = 20$ (Answer 2 out of 3)

- What is β – elimination? Explain the mechanism and orientation of E1 reaction. Compare between E1 and E2 reaction.
- Explain the mechanism and stereochemistry of SN_2 reaction. Add a note on factors affecting SN_2 reaction.
- Explain, why Aldehydes and ketones are susceptible for nucleophilic addition reactions? Give any two methods for preparation of aldehydes. Write a note on Perkin reaction.

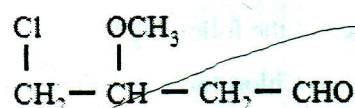
Q. 3. Answer the following questions. = $7 \times 5 = 35$ (Answer 7 out of 9)

- i) Explain the different types of structural isomerism with examples.
- ii) Explain the mechanism of halogenation of alkanes.
- iii) Explain the qualitative tests for alcohols.
- iv) Explain the addition of water and alcohol to aldehydes.
- v) Arrange the following compounds in order of increasing acidity and justify your answer.
Chloroacetic acid, formic acid, acetic acid, dichloroacetic acid.
- vi) Give the structure and uses of acetic acid, lactic acid, formaldehyde, glycerol, ethyl alcohol and dichloromethane
- vii) Explain why aliphatic amines are more basic than ammonia.
- viii) Write a note on Ozonolysis and Diel-alder reaction.
- ix) Give the IUPAC names of following compounds.

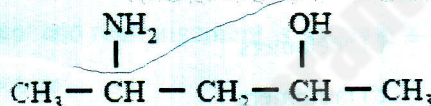
a)



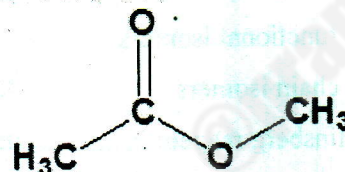
b)



c)



d)



e)



-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 09/05/2023

Course: B. Pharmacy

Subject Name: Pharmaceutical Organic Chemistry-I

Max Marks: 75

Sem: II

Subject Code: BP202T

Duration: 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Give structure and uses of Ethanolamine and Benzoic acid.
- ii) Enlist any two qualitative tests for carboxylic acids.
- iii) What is saytzeff's rule? Give its reaction.
- iv) Write in short stability of alkene.
- v) Define functional isomerism with example.
- vi) Give two examples for nucleophilic addition reaction to carbonyl compounds.
- vii) Draw the structure and write uses of Iodoform.
- viii) Draw structure and write uses of Propylene glycol.
- ix) Draw the structure and give IUPAC name of isobutanol.
- x) Enlist any four reactions of alkyl halide.

Q. 2. Long Answers (Answer 2 out of 3)

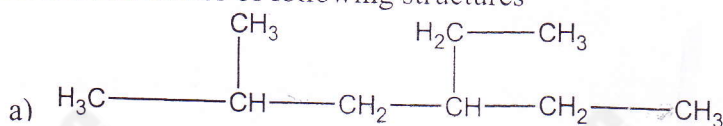
(2 x 10) = 20

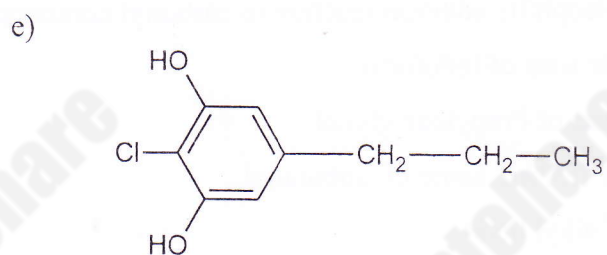
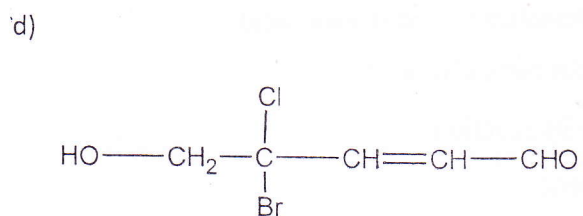
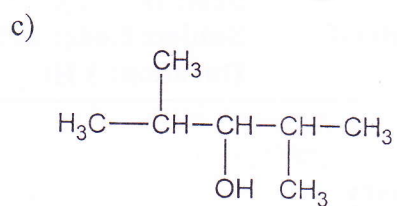
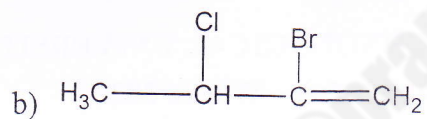
- i) Illustrate the mechanism of Markownikoff's and anti- Markownikoff's orientation of alkene with suitable examples.
- ii) Explain the reaction and mechanism of Perkin and Aldol condensation.
- iii) Explain sp^2 hybridization in alkenes. Write a note on Ozonolysis and Diels-alder reaction.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Explain the sp^3 hybridization in alkanes with suitable example.
- ii) Classify organic compounds with example.
- iii) What is effect of substitution on acidity of carboxylic acid?
- iv) Explain the factors affecting $E1$ reaction.
- v) Give an account on Cannizzaro and cross-cannizzaro reaction.
- vi) How you will identify alcohol and aliphatic amines by qualitative tests?
- vii) Discuss the free radical addition reactions of conjugated dienes.
- viii) How you will synthesize alkenes?
- ix) Write IUPAC names of following structures





-----END OF THE PAPER-----



Course : B. Pharmacy

Date: 08- 09-2023

Subject Name : Pharmaceutical Organic Chemistry-I

Sem:

II

Subject Code :

BP202T

Max Marks : 75

Duration :

3 Hr.

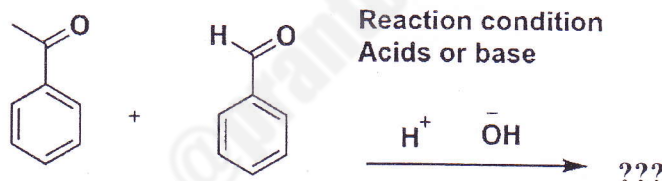
Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Define electrophile and nucleophile with example?
- ii) Write down the structures of
 - a) 2-methyl-3-chloro-1-pentene
 - b) 5-fluoro-3-hydroxy-pentanoic acid
- iii) Enlist the qualitative test for Aldehyde?
- iv) Explain basicity of amine?
- v) Write down the structure and uses of benzyl alcohol?
- vi) Enlist the reactions of carbonyl compounds?
- vii) What is the hybridization of carbon atom present in
 - a) Alkanes
 - b) Alkenes
- viii) Predict the product as per the Crossed Aldol Condensation in the following reaction?



- ix) Define carbocation and carbanion?
- x) What is inductive effect?

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) What elimination reaction? Explain reaction, mechanism kinetics, and order of reactivity of E1 and E2 reaction?
- ii) Why Aldehydes and ketones are susceptible for nucleophilic addition reactions? Explain the reaction and mechanism of Aldol condensation and benzoin condensation?
- iii) Write in detail about bimolecular nucleophilic substitution reaction (SN2) along with suitable example?

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Write down the Rules for IUPAC Nomenclature for organic compounds?
- ii) Write a short note on isomerism in organic compounds?
- iii) Discuss about the free radical addition reactions of conjugated dienes?
- iv) Describe with example, halogenation of alkane?
- v) Explain Diel-Alder reaction?
- vi) Write short note on Cannizzaro's reaction?
- vii) Write about the Markownikoff's rule and Anti-Markownikoff's rule?
- viii) What are carboxylic acids? Explain the effect of substituents on acidity of carboxylic acids?
- ix) Draw the structures of a) Benzoic acid b) Ethanolamine c) Ethylenediamine d) Dichloromethane e) Benzaldehyde



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Summer Examination – 2024

Course: B. Pharmacy

Semester: II

Subject Name: Pharmaceutical Organic Chemistry-I

Subject Code: BP202T

Max Marks: 75

Date: 14-06-2024

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1.	Objective Type Questions (Answer all the questions)	(10 X 2) = 20
i)	Describe any two qualitative tests for carboxylic acid.	
ii)	Define functional isomerism with example.	
iii)	Explain Inductive effect.	
iv)	Write a note on ozonolysis reaction.	
v)	Write a difference between E ₁ and E ₂ reaction.	
vi)	What is meant by Electromeric effect?	
vii)	Draw the structure and uses of Ethyl alcohol.	
viii)	Write the IUPAC names of the following compounds.	
	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C}-\text{CH}-\text{CH}-\text{CH}-\text{CH}_3 \\ \quad \\ \text{CH}_3 \quad \text{CH}_3 \end{array}$ a) $\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3-\text{C}-\text{CH}-\text{C}-\text{H} \\ \quad \\ \text{O} \quad \text{O} \end{array}$ b)	
ix)	Draw the structure of the following compounds.	
	a) 2-buten-1-ol b) 4-hydroxy pentanoic acid	
x)	Write is saytzeff's rule? Give its reaction.	
Q.2.	Long Answers (Answer 2 out of 3)	(10 X 2) = 20
i)	What is elimination reaction? Explain reaction, mechanism, kinetics and order of reactivity of E ₁ and E ₂ reaction?	
ii)	Illustrate the reaction and mechanism of Markownikoff's and anti-Markownikoff's orientation of alkene with suitable example.	
iii)	Explain the reaction, mechanism and stereochemistry of S _N 2 reaction	
Q.3.	Short Answers (Answer 7 out of 9)	(5 X 7) = 35
i)	Explain the sp ² hybridization in alkanes with suitable example.	
ii)	Discuss the free radical addition reactions of conjugated dienes.	
iii)	Write a short note on Cannizzaro's reaction.	
iv)	What are carboxylic acids? Explain the effect of the substituents on acidity of carboxylic acid?	

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Branch-Pharmacy

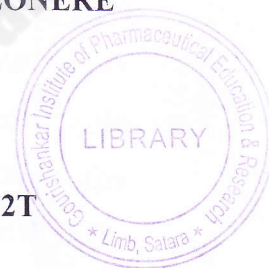
Supplementary End Semester Examination – Winter 2023

Date: 05/01/2024

Course: B. Pharmacy
Subject Name: Pharmaceutical Organic Chemistry- I

Sem: II
Subject Code: BP202T
Duration: 3 Hrs

Max Marks: 75



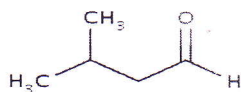
Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Draw the following structures
1) 1-Methyl cyclopentane 2) 5-Ethyl-3-methyloctane
- ii) Define the following terms
1) Hybridisation 2) sp^2 hybridisation
- iii) Identify the following structure



- iv) Write on Ozonolysis reaction.
- v) Write difference between E_1 and E_2 reactions.
- vi) Enlist any two qualitative tests for carboxylic acids.
- vii) Draw the structure and write uses of Ethanolamine.
- viii) What is mean by Electromeric effect?
- ix) Draw the structure and write uses of Ethyl alcohol.
- x) What is Saytzeff's rule? Give its reaction.

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) Describe the SN_1 & SN_2 reactions in detail.
- ii) Explain in detail Aldol condensation and Cannizarao reaction.
- iii) Illustrate the mechanism of Markownikoff's and anti- Markownikoff's rule. And write on orientation of alkene with suitable examples.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Describe the Perkin condensation reaction.
- ii) Elaborate sp^3 hybridisation in alkane.
- iii) Write different chemical reactions shown by alkyl halide.

- iv) What are Diene? Classify with example.
- v) Write a note on Inductive effect.
- vi) Explain acidity of carboxylic acids with effect of substituents on acidity.
- vii) What is structural isomerism in organic compound?
- viii) Discuss on Benzoin condensation reaction.
- ix) What is mean by aliphatic amines and explain basicity and effect of substituent on basicity?

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